Dr. B. M. PRASANNA
Associate Professor
Department of Chemistry
Bapuji Institute of Engineering and Technology
DAVANAGERE - 577 004.

E - Waste Management

Introduction: In the new world of materials, usage of electrical and electronic items has been increasing rapidly year by year. These materials have lesser life span and also used for lesser duration due to fast change in feautures and the capabilities. cumulative result of increased manufacturing, usage and discarding is rapid increase the amount of end of life electronics and electrical items. All electronic and electrical items which are discarded on completion of their useful life together is called e-waste. There is rapid growth in e-waste generation since 1990s and will continue at faster rate in future.

As per an estimate, the average annual global E-waste volume is about 65.4 million tons in the year 2017.

Dr. B. M. PRASANNA
Associate Professor
Department of Chemistry
Bapuji Institute of Engineering and Technology
DAVANAGERE - 577 004.

Sources of E-waste:

main sources of e-waste are,

1) computer peripherals : Monitor, Key board, mouse, mother board, Laptops, CDs etcl-
2) Telecommunication Devices: phones, cell phones, routers, Pagers, fax machine etcl-.
3) House hold appliances : TV, fridge, washing machine, Video players, ovens etcl-.
4) Industrial electronics: sensors, medical devices, automobile devices etcl-.
5) Electrical devices : switches, wires, Bulbs etcl-

Composition of E-waste:

E-waste is characterised by morethan 1000 hazardous and non-hozordous materials.

1) E-waste contains about 65 % of iron, steel and other metallic materials including costly metals like platinum, gold, Silver, and toxic

metals like lead, mercury, cadmium, chromium etc.-.

2) E-waste contains about 21% of polymeric non-biodegradable materials including poly-vinyl chloride (PVCs), polychlorinated biphenyls and brominated flame retardant plastics.

3) E-waste also contains about 11.8% of CRT and LCD screens and other materials like glass, wood, play wood and ceramics.

most of these products are ended up in landfills, rubbish dumps, and recycling centres creating several complications for waste management obbicials, policy makers, and residents.

Characterstics of E-waste :

Need of E-waste Management:

E-waste is a complex mixture of metals, organics and ceramics. Disposing of this mixture in landfills can cause serious environmental issues. The amounts being buried have been increasing in the landfill sites. This results in serious environmental problems due to passing of dangerous chemicals into soil, water, air and human beings. Also, it results in loss of metals whose quantities are limited in the earth's crust. Due to these ~~effects~~ facts, proper efficient e-waste management is required for the recovery and reuse of components of e-waste.

Toxic Materials used in manufacturing Electronic and electrical products.

According to toxicity, the types of substances in E-waste can be classified into two types,

1) Hazardous substances.
2) Non-Hazardous substances.

Hazardous substances are toxic, and can affect the quality of the ecosystem and can have determental effects on human health. These includes,

1) Heavy metals like cd, Cr, Pb and Hg.

2) organic compounds like chlorofluoro carbon. polycyclic aromatic hydrocarbons (PAHs), polybrominated diphenyl ethers (PBDEs) polychlorinated biphenyls (PCBs). and polychlorinated dibenzo-dioxin furans (PCDD/Fs)

Health Hazards due to exposure to E-waste

Major problem with the e-waste is passing of toxic metals and organic compounds present in e-waste in to ecosystem that is air, water ~~and~~ soil and ultimately to living beings. contamination of ecosystem by e-waste hazards is mainly due to,

1) Informal e-waste recycling techniques employed for metal recovery from printed circuit boards. Although the recycling is responsible to remove

or delay the release of contaminants into the environment. unscientific ways used for recycling may result in accumulation in landfills and end up within the environment.

2) Open burning of printed circuit boards and electric cables to recover copper. This ~~relates~~ releases doxins, persistent organic pollutants, PAHs, PCBs, persistent halogenated compounds into the environment.

3) unscientific dumping of e-waste. The effluent from e-waste recycling centres and the dumping sites are rich with heavy metals and suspended particulate matters.

4) The use of primitive methods to process e-waste has resulted in contamination of the soil. Many of the effluents are acidic and responsible for changing the pH of the natural soil and results in reduction of harvest in commercial agricultural systems. concentration of Pb, Ba, Cd, Hg, Cr and Zn is found to be very much higher near dumping sites.

Toxic elements were found to disturb the nitrogen and potassium absorption by plants which are important component in plant growth and developement. Toxic substances gradually through crops enter into ecosystem.

5) waste water from e-waste recycle or dumping sites is another major tranport pathway for contamination of aquatic systems. open burning sites of e-waste for metal recovery is the major contributor for release of different contaminants from e-waste to atmosphere. Toxic substances are entering human body from air, water and soil through inhalation, ingestion and dermal contact pathways. They are having different types of serious health impacts.

6) Most of these heavy metals are carcinogenic. Particulate maters released from open burning can adsorb metal particles and enter into human body.

7) people living near to e-waste collection, dumping, and incineration sites are direct victims for the toxic contaminants. There are higher number of incidences with skin damages, headache, vertigo, chronic gastrics, gastric ulcers & duodenal ulcers.

8). contaminants such as heavy metals, doxins, furans, PAHs, PCBs and polybrominated diphenyls are able to create human gastrointestinal irritation and laxative effects, ~~abnormal~~ ~~sper~~ chromosome aberration, DNA damage, reduced fecundity and adverse birth effects.

9) cadmium (Cd) metal can damage kidneys and lead (Pb) metal has a direct influence toward central nervous system as well as deterioration of intelligent quotient in childrens. mercury is a potential mutagenic compound and can greatly affect neurons.

10) placentas collected from mothers who live near e-waste dumping sites have shown very high concentration of toxic chemicals.

11) Different hazardous chemicals can accumulate in human blood, serum and urinary tract and finally show synergetic effects with a cumulative impact.

Recycling & Recovery

In e-waste, among various constituents, metal contribute to the significant economic value, and the efforts are focoused on extracting those metals during recycling operations.

Recovery and Recycling Technologies:

The recycling of e-waste is intiated with physical or mechanical pre-processes.

I step) Pre-treatment step involves physical removal of toxic materials and unwanted components and then manual dismantling and seperation of components such as PCBs, monitors, batteries etc/-. into various fractions like metals, ceramics plastics, wood and paper using hammers, screwdrivers and conveyer beds for disassembling.

II-step: Shredding of the materials mechanically through crushers and grinders to collect fragments of metal bearing components.

III-step: The waste is passed through electrical seperator to seperate the metallic and non-metallic components.

and magnetic seperator is used to seperate ferrous metals. Gravity seperation is used to seperate Al. metal.

IV step: After physical seperation various chemical techniques are employed to recover metals.

Metals can be recycled any number of times and it contributes to conservation of the natural resources. And also recovery of metals from e-waste is more economical than extraction from their ores. e-waste contains 10 times more excessive concentration of gold compared to gold ores. concentration of ~~gold~~ copper in PCB is 20-40 times more than that is present in its ore.

Most widely used recovery techniques are,

1) Pyrometallurgical process

2) Hydrometallurgical process.

Dr. B. M. PRASANNA
Associate Professor
Department of Chemistry
Bapuji Institute of Engineering and Technology
DAVANAGERE - 577 004

Pyrometallurgical process

Pyrometallurgy technology is used to extract pure non-ferrous and precious metals from e-waste. The methods require elevated temperatures to reduce or extract metals and therefore require high amount of energy input.

Smelting, combustion, pyrolysis and molten salt processes are the main pyrometallurgical methods employed for recovery of metals from e-waste.

(a) Smelting: copper smelting is commonly used for recovery of nonferrous metal fractions from e-waste. The processed scrap after preliminary stage contains mainly iron (Fe), Aluminium (Al), copper (Cu), lead (Pb), tin (Sn), antimony (Sb), Zinc (Zn) and preceious metals as metallic constituents..

This mixture is fed into copper smelters. During smelting Pb, Sn, Sb and refined by electrometallurgy. Here the anode is dissolved and pure copper (99.99%) is deposited over cathode leaving a slury ~~sludge~~ residue.. This slury residue is called anodic slime and contains undissolved metallic fractions.

(b) combustion of e-waste:

combustion is a low-technology, low cost, straight farward operation focousing only on the recovery of valuable metals. Here e-waste is subjected to open burning in uncontrolled manner which releases all sorts of pollutants into atmosphere. Hence the method is highly dangerous for the environment and also increases the health risk of the workers

(c) Incineration of e-waste:

Incineration is a controlled combustion of the waste with suitiable emission units. The incinerator has two connected furnaces. In the first furnace, e-waste is burnt at temperature above 800 °c and in the second stage for the gaseous products of the first incinerator are further oxidised above 1100 °c. Heavy metals are collected in the bottom and fly ash. Hydrometallurgical processes are used for further recovery of metals.

pyrolysis of e-waste : pyrolysis is a thermal decomposition of e-waste ~~with suitable emission units~~ at higher temperature in an oxygen free environment. During pyrolysis, irreversible thermal decomposition reactions takes place, leading to the formation of lower-molecular weight products (gas & liquids) at temperatures between 450 & 1100 °C. pyrolysis gases, oils and char have an economical value and can be used as fuel and chemical feedstock. The metallic components can be recovered easily by seperation. Main danger with this process is release of toxic halogens into atmosphere with flue gases.

(c) Molten salt process :

In this process, the e-waste is feed in with the salt, and salt is melted at the desired temperature under an inert atmosphere environment. The organic parts decompose in the molten salt forming carbonates and silicates and are converted to alkali metal halides, which remain in the molten salt. Therefore, the flue gas contains a high content of $H_2(g)$ that can be stored as a fuel. The metallic component is liberated by removing the organics, and is ~~called~~ collected on the bottom of the furnace. After removing molten salt, mixture containing valuable metals is further treated to obtain pure metals. Various molten mixture, are used at different temperatures between 300 and 1100 °C depending on the requirement.

Hydrometallurgical process:

Traditional hydrometallurgical process which have been used from so many years for metal extraction from mineral ores are used for the recovery of metals from e-waste.

There are three stages in metal recovery by hydrothermal method,

(a) pre-treatment stage.

(b) chemical treatment stage.

(c) Metal recovery stage.

a) pre-treatment stage:

pre-treatment stage involves physical seperation of metal components from e-waste as discussed above followed by smelting of the mixture in some cases. In hydrometullurgical process, main steps are chemical treatment and metal recovery steps.

In chemical treatment stage, the metals are made to leach into solutions using different leaching reagents. In the metal recovery stage, metals are recovered from the leached solutions using through electro-refining precipitation, cementation adsorption, solvent extraction, and ion exchange methods. ~~These techniques are briefly given below,~~

(b) Chemical Treatment Stage:

(1) cyaninde leaching: Eventhough, cyanide solutions are toxic, they are mainly used to leach gold metal. Alkali cyanide like KCN with 3-nitrobenzene ~~sulfonic~~ sulphonic acid sodium salts are used to dissolve gold under aerated condition. Further, the same solution can be subjected to electroplating

to obtain pure gold metal. copper also re
dissolves in cyanide solutions.

(2) Acid & Alkali leaching : Nitric acids, sulfuric acids and hydrochloric acids are the prominient mineral acids used for leaching of targeted metals. A few organic acids such as ascorbic acids, citric acids and acetic acids are also used to leach light metals from spent batteries and mobile devices. Li is leached from lithium ion battery ~~battery~~ waste by treatment with citric acids and hydrogen peroxide as reducing agent.

(3) Thiosulphate Leaching: Thiosulphate is used an alternative leaching agent ~~to~~ for the recovery of precious metals like gold and silver. solubilizing effect of thiosulphate reagent is better and its intereference with the other cationic species is also very less. Ammonium thiosulfate solution is used to solubilize gold, silver, platinum, and other precious metals in the form of anionic stable complexes for a wide range of pH.

This reagent is safe, non-toxic, non-corrosive and metal can be recovered readily from the complex.

4) Thiourea Leaching: Thiourea is a sulfur-based organic complexing agent, that forms a cationic soluble complex with the target metal. Thiourea give quick rates of leaching. less intereference of ions. is environmentally friendly, and has a low cost. The leaching rates are up to 99% for this complex. Thiourea is not stable and decomposes easily in alkali environment, hence the reaction requires an acidic media. In printed circuit boards, the gold and silver selectively forms a metal-thiourea cationic complex.

5) Halide Leaching: chloride, bromide and iodide ions can be used to leach gold from the PCB waste. They are employed as a replacement for cyanide leaching agents. They exhibit in high solubility, improved redox potentials, and high rates of leaching. They are ~~chep~~ cheaper, selective to the target, and ideal leaching agent.

(C) Metal Recovery stage:

In this stage, pure metal is recovered from leached metal solution. A variety of methods such as solvent extraction, ion exchange, adsorption, precipitation, and cementation are available. Method employed depends upon the nature of leached solution. In majority cases, metals can be recovered selectively.

~~(4)~~ (1) Solvent extraction: Leaching solution is treated with an organic solvent in a seperating funnel. It results in two phase system. Metal is extracted from leached solution phase to organic phase. Different extractants such as ~~ionic~~, anionic, cationic or solvating type are employed. The anionic type like amides or amines are used for the extraction of vannadium, gold, iridium and sometimes rhodium and tungsten. Methyl - iso - butyl ketone is used as an extractant for gold. Amide extractant is used for chloride leached iridium found in electronic waste. Diamine extracts are used for platinum and palladium.

(b) Electrodeposition: In this technique, pure metal is obtained from leached solution by electrodeposition. pure metal same as metal to be extracted is taken as cathode and inert metal is used as anode. They are dipped in leaching solution. when current is applied, pure metal is electrodeposited on cathode.

Electrodeposition techniques have the advantages of high efficiency, less use of auxiliary materials, low environmental impact, and cost effectiveness. Lead, Tin, and copper from PCBs can be recovered by leaching followed by electrodeposition. with appropriate planning, both leaching step and recovery step can be clubbed together. For example, copper can be leached from PCBs with simultaneous cathodic electrodeposition from the leaching solution with 99.95 % efficiency leaving the residue rich in gold metal.

(c) Ion exchange: This is an improved version of solvent extraction method. Here solvent extracting reagents are impregnated on polymer ~~beds~~ beads.

The functional group of reagents acts as chelating group and selectively bind to metal ions. Thus they can be used for selective recovery of the desired metal ion. Some advantages of using ion exchange resins for recovery of metals are their ease to use, no loss of reagent, no seperation of phases, low cost, use in low concentrations of metal ions and environmental saftey.

This method was used to recover uranium, Palladium group metals in ion exchange resins. In aqueous solutions, the cation exchange resin KB-2 and the anion exchange resin AN-108, synthesized with long chain cross-linking agents can completely remove Mn (II) and Cr (VI) respectively. Ion exchange resins have found to be effective in recovering gold from cyanide and thiosulfate leach solutions.

(d) Adsorption: Metals can be recovered from leached solutions by adsorption on appropriate adsorbents. Activated carbon is found to be effective adsorbent. The adsorption of gold on activated carbon from cyanide solution is an efficient, cost-effective process. It is commercially called as carbon-in-leach (CIL) technology.

Adsorption of gold-thiourea complex on activated carbon is also effective. Pt, Au, Ag and Cu metals were removed from their cyanide solutions with 95-100 % efficiency.

Extraction of Gold from E-waste:

Gold metal has good electric conductivity and chemical stability and hence it is used for making integrated circuits of electronic devices, coating for contacts and connectors.

In e-waste, PCBs are rich in metals. It contains around 35 % Cu, 0.16 % Silver and 0.13 % gold by weight. Several techniques such as pyrometallurgy, Hydrometallurgy, microwave treatment and plasma technology are employed to recover precious metals from e-waste. Among these, recovery of metals using hydrometallurgical route is more economical.

The hydrometullurgical Method:

There are three stages of hydro-metullurgical processes are,

1) pre-treatment stage.
2) chemical-treatment stage.
3) Metal recovery stage.

1) **pre-treatment Stage:** In this stage, e-waste is mannually dismantled to seperate various fractions like metals, ceramics, plastics, wood, and paper. Techniques such as gravity seperation, electrostatic seperation, magnetic seperation and eddy current seperation are used to seperate metals from other fractions.

2) **Chemical Tretment Stage:** In this stage, targetted metals are leached into solution by treating with appropriate chemical reagents. several leaching agents such as thiosulfate, alkali, ~~cyanidi~~ cyanide and many acids

uch as hydrochloric acid, sulfuric acid, and nitric acid can be used to leach gold in the solutions.

cyanide leaching is the most common method used to extract gold metal. Sodium salt of 3-nitrobenzene sulfonic acid with potassium cyanide (KCN) in the presence of oxygen is used as leaching agent. A water soluable dicyanoaurate gold complex is obtained in the process.

$$4\,Au + 8CN^{\ominus} + O_2 + 2H_2O \longrightarrow 4\,Au(CN)_2^- + 4(OH)^{\ominus}$$

Metal Recovery stage:

In the last step, metal is recovered from leach solutions. varieties of methods like electrodeposition, solvent extraction, ion exchange, adsorption, precipitation and cementation are used to recover metals from leached solutions. selective recovery is possible for most processes.

Gold can also be extracted from leaching solution by electrodeposition of gold from dicyanoaurate gold complex. pure gold metal taken as cathode, and inert anode are dipped in leached solution. when current is applied, gold is electrodeposited on cathode.

Direct Recycling of E-waste:

~~The techniques are used to recover useful chemical components from~~

The techniques are used to prevent disposal of e-waste by means of dumping or landfilling. Because, this results in waste of valuable materials and can also cause environmental problem due to contamination by heavy metals. But all those processes require lot of energy and money. Another cost effective method for recycling of e-waste is by direct recycling.

Direct recycling means harvesting electronic components directly from e-waste without breaking them further into small components. Harvested materials are further processed with healing methods to regenerate recycled materials.

The regenerated materials have performance equivalent to originally manufactured materials. Thus in this method, all the complicated chemical and metallurgical steps involved in conversion of e-waste components into original chemicals are avoided. Recycling loop is shortened. Hence, it requires less amounts of energy and reagents and hence it is the most enviromentally friendly method. However, there are few problems associated with direct recycling method. Direct regeneration of components depends on state-of-health of used electronic materials. Defects and impurities accumulated during usage could affect the quality of the refurbished active materials. And also direct recycling may not restore the intial properties of pristine active materials.

Presently, direct recycling of lithium ion batteries is studied extensively. Here the battery is dishanged first to avoid short circuiting and self-ignition of battery and dismantled further, but each component is regenerated by appropriate process to recover its function. These components are reassembled for reuse.

Role of Stake Holders in Environmental Management of E-waste (producers, consumers, Recyclers, and statutory Bodies)

Basically there are four stakeholders in environmental management of the e-waste.

They are

1) statutory Government Regulatory Bodies.

2) producers (Manufacturing units)

3) Recyclers

4) consumers

The E-waste management program is designed by statutory Government regulatory bodies. The members of the body frame the policies and execute it for protection of environment. To achieve the plan of management of e-waste a green tax is collected from consumer through manufacturer.

penalties are implied on manufacturer and consumers when green tax is not paid. Manufacturing units must support the agenda of e-waste management by doing dismantling, processing of e-wastes, managements of scrap materials and reselling of recycled materials. The presence of collection units for door to door collection of e-waste. consumer must pay green tax and be aware of importance of e-waste management. All four stake holders must work in tandem form effective e-waste management.

(1) Statutory Government Regulatory Bodies:

The statutory bodies play a pivotal role in management of e-waste. Main roles of statutory bodies are,

(a) To collect green tax from consumer through producers.

(b) Apply some extra charges on the producers (Manufacturing units) in the form of penalty when no proper recycling of e-waste is assured from manufacturing units.

(c) provide the incentives in the form of subsidy to recyclers and collectors when recycling of e-waste is monitored properly.

(d) To conduct the programs of awarness in the society about importance of e-waste recycling in reduction of hazardous substances.

2) producers (Manufacturing units) :

The main role of producers in management of e-waste are,

(a) The accountability of collect green tax.

(b) Charging an additional amount on consumer during sell of e-products and returning it with interest at the time of exchange of e-product.

(c) Forming the ~~grp~~ group of manufactures who monitor and encourage the recycling of e-waste.

(d) Bearing the transportation cost and collection fees to ease the collection process.

(e) purchase the recycled material at ~~the~~ fixed value and using of recycled e-waste during manufacturing.

f) Giving discount to consumer on the basis of e-waste generated from gadgets.

3) Recyclers (Recycling & collection units)

The main roles of recyclers in the management of e-waste are,

(a) The accountability of recycling units is dismantling, recycling processing of e-waste materials. Management of scrap materials (like incineration) and resulting of recycled materials.

(b) Establish the collection units and the group of people who can ensure return back of e-products by consumer in exchange offer or directly approach consumer for door to door collection.

(c) collect the e-waste from the collection units, dealer or retailer.

(d) providing incentives when proper collection of e-waste assured by collection units.

4) <u>Consumer</u>: The main roles of consumer in management of e-waste are,

(a) The accountability to pay green tax.

(b) Develop self-awarness on e-waste management and involve in awarness programms.

(C) Returning back of e-waste to collection units.

~o~